C

PROJECT DOCUMENTATION

ChessMaster

A browser-based chess game with AI and local play, a responsive game dashboard, practice positions, and opening study notes.

Open the live game

Application guide · GitHub Pages edition

1. Project overview

ChessMaster is a client-side chess application built with Next.js, React, TypeScript, chess.js, and react-chessboard. It provides a playable chessboard, computer opponent, local two-player mode, move controls, timers, and supporting study and profile panels.

The deployed site is hosted on GitHub Pages as a static export. It does not use a server, database, or external authentication provider.

2. Features

Area	What it provides
Chess play	Interactive 8×8 board; click-to-move and drag-and-drop; legal move highlighting; board flip; promotion selection; move list and game status.
Rules	Move legality and standard chess rule state are handled by chess.js, including check, checkmate, stalemate, castling, en passant, promotion, repetition, and insufficient material.
Game modes	Play against the built-in AI or pass the same device between two local players.
AI difficulty	Beginner, Intermediate, Advanced, and Expert settings. The built-in opponent uses a small minimax search with alpha-beta pruning and a basic material/position evaluation.
Game tools	Undo, restart, 10-minute-per-side timer, optional synthesized move/capture sounds, move history, PGN preview, and copy-to-clipboard export.
Practice and study	Preset challenge positions and concise opening lesson cards.
Player dashboard	Local profile, rating-like score, recent results, win/loss/draw counts, win-rate display, and locally collected ranking list.
Responsive UI	Dark navy and gold visual theme with layouts that adapt to desktop and smaller screens.

3. Playing a game

1. Open the live game or start it locally using the instructions below.
2. Register a local profile or sign in to a profile stored in this browser.
3. Choose **Vs AI** or **Local**. Select an AI difficulty for computer games.

4. Move by selecting a piece and destination square, or drag a piece to a legal square. Legal destinations are highlighted when a piece is selected.
5. Choose a promotion piece when a pawn reaches the final rank.
6. Use **Undo**, **Restart**, **Flip Board**, and **Export PGN** as needed. The sound toggle controls browser-generated move sounds.

> **Timer:** Both sides begin with 10 minutes. The timer starts after a profile is available and runs during an unfinished game.

4. Local setup

Requirements

- Node.js 20 or later (the Pages workflow uses Node.js 20).
- npm.

Run the development server

```
git clone https://github.com/isaacwatt40-commits/chessmaster.git
cd chessmaster
npm ci
npm run dev
```

Open `http://localhost:3000` in a modern browser. The development server watches source changes and refreshes the app.

Build and lint

```
npm run lint
npm run build
```

The build creates the static site in the `out/` directory because the project is configured for Next.js static export.

5. GitHub Pages deployment

The production site is published at https://isaacwatt40-commits.github.io/chessmaster/.

- Workflow: `.github/workflows/deploy.yml`.
- Trigger: push to the `master` branch or manual workflow dispatch.
- The workflow installs locked dependencies with `npm ci`, builds using the repository base path, uploads `out/`, and deploys with GitHub Pages.
- GitHub Pages is configured to use the GitHub Actions deployment source.

To publish a change, commit and push it to `master`. Then check the repository's Actions tab for a successful "Deploy ChessMaster to GitHub Pages" run. GitHub Pages may take a short time to update after the workflow completes.

```
git add .
git commit -m "Describe your change"
git push origin master
```

6. Technology and source layout

Path	Purpose
`src/app/page.tsx`	Main interactive chess page, game state, AI opponent, controls, profile UI, dashboard and study panels.
`src/app/globals.css`	Global CSS and Tailwind import.
`src/app/layout.tsx`	Root layout, fonts, and page metadata.
`next.config.ts`	Next.js static export and GitHub Pages base-path configuration.
`.github/workflows/deploy.yml`	Automated build and GitHub Pages deployment.
`package.json`	Dependencies and development, lint, and build scripts.

Core libraries include Next.js 16, React 19, TypeScript, chess.js, react-chessboard, lucide-react, and Tailwind CSS 4.

7. Data, privacy, and current limitations

- **Profile and results are browser-local.** Profile, dashboard, ranking, and match-history data use localStorage in the current browser. They do not sync across devices and can be removed by clearing browser storage.
- **Authentication is a prototype UI.** The sign-in/register form does not verify passwords or provide secure authentication. Do not use real or sensitive passwords in it.
- **No backend or cloud database.** The static Pages deployment cannot provide server-side accounts, persistent shared rankings, or server-managed game records.
- **No online multiplayer.** Local mode is pass-and-play on one device; there is no lobby or network matchmaking.
- **AI is not Stockfish.** It is a lightweight in-browser search opponent. Advanced and Expert currently use the same search depth and it is not intended to match a dedicated engine.

- **Study content is introductory.** Challenge positions and opening lessons are preset client-side content; they are not a dynamically updated daily puzzle service or a full interactive course.
- **Timer settings are fixed.** The current game uses a 10-minute starting time for each side; custom time controls are not provided.

For production accounts, cross-device records, online play, or authoritative rankings, add a secure backend and authentication service (for example, Supabase Auth and a database) and review data protection requirements before collecting personal information.

8. Troubleshooting

- **Local site does not open:** confirm the dev server is still running in the terminal, then use `http://localhost:3000`.
- **Profile or statistics disappeared:** browser storage may have been cleared, or the site may be open under a different browser/profile.
- **GitHub Pages has not updated:** inspect the latest Actions run. A deployment only publishes if both build and deploy jobs succeed.
- **PGN copy does not work:** clipboard access may require a secure context or browser permission. The PGN preview remains available in the app to select and copy manually.
- **No move sound:** confirm sound is enabled and that the browser permits audio playback after user interaction.

ChessMaster project guide · Repository: github.com/isaacwatt40-commits/chessmaster · Live site: isaacwatt40-commits.github.io/chessmaster/